

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2023-2024
FULL MARKS: 75

SWE 4401: Software Requirement and Specifications

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

Case Study: The "UniConnect" Mobile App

A university wants to build a new mobile application called UniConnect. The primary goal of the app is to increase student engagement by making it easier for students to discover, join, and participate in campus clubs. It should also simplify the administrative tasks for club leaders.

1. You are a business analyst on the UniConnect project. You have just interviewed Farah Hossain, a busy third-year student and the President of the University Debating Society. Below is a transcript of her key points.

"Honestly, managing the club is like a second job. At the start of every semester, we have a booth at the activities fair and get hundreds of students signing up on paper forms. I then spend days manually typing their names and emails into a spreadsheet. It is so easy to make a typo, and then they never get our communications. We need a way for students to just sign up directly. My biggest headache is communication; I send out weekly emails about meetings and events, but students complain that they get lost in their crowded inboxes. I wish I could just send a quick announcement directly to their phones. That would save me so much time and ensure everyone knows what's happening."

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| a) Formulate a detailed User Persona for Farah Hossain based on the interview transcript. | 9
(CO2)
(PO3) |
| b) Construct three (3) distinct and well-formed User Stories for the UniConnect app that address Farah's needs. | 6
(CO1)
(PO2) |
| c) Develop a set of Acceptance Criteria for one of the user stories you constructed in part (b) that relates to communication. Use the Gherkin format to create one "Happy Path" scenario and one "Unhappy Path" scenario. | 10
(CO1)
(PO1) |
2. The development team has created a "first draft" of a requirements list based on early discussions. The list below contains several poorly formed requirements.
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|---|-------------------------|
| i. The app must be fast and user-friendly. | 5 × 6
(CO1)
(PO2) |
| ii. As a user, I want to manage my club profile. | |
| iii. The system shall be secure. | |
| iv. As a club president, I want a button to approve new members. | |
| v. The app will use a PostgreSQL database and a React Native front-end. | |

For each of the five (5) requirements in the list above, you must (a) analyze the requirement to identify its primary flaw and explain why it is weak, using specific terminology from the course. Then (b) propose a superior, well-formed alternative that is specific, testable, and adheres to the principles of good requirement writing.

3. After a refinement session, the team has a clean, well-written backlog of user stories for the first major release (MVP) of UniConnect.

Cleaned Backlog:

- US-01: As a student, I want to search for clubs by name or category, so that I can find groups that match my interests.
- US-02: As a student, I want to view a club's profile page with its description and meeting times, so that I can learn more about it.
- US-03: As a student, I want to join a club by tapping a "Join" button on its profile page, so that I can become a member. ✓
- US-04: As a club president, I want to view a list of pending membership requests, so that I can manage who joins my club. ✓
- US-05: As a club president, I want to approve or deny membership requests, so that I can finalize my member list. ✓
- US-06: As a student, I want to receive push notifications for announcements from clubs I have joined, so that I stay informed.
- US-07: As a student, I want to customize my personal profile with a photo and a short bio, so that I can personalize my identity in the app. ✓

- a) Apply the MoSCoW method to classify six (6) user stories from the list into the categories Must Have, Should Have, and Could Have. You must classify at least one story in each category. Justify your classification for each story with a brief (1-2 sentence) rationale. 12 (CO1) (PO1)
- b) Explain the relationships between different levels of requirements and agile processes. 8 (CO1) (PO1)
 - i. The high-level Business Requirement for UniConnect is "to increase student engagement." Illustrate how a specific Functional Requirement (from the backlog) and a Non-Functional Requirement (e.g., related to performance or usability) both trace back to and support this business-level objective.
 - ii. Identify the specific Scrum event where the prioritization activity from part (a) would most likely occur. Name the role primarily responsible for these prioritization decisions.